# Roberto Rocchetta

Curriculum Vitae | Research Leadership, Publications, Funding, and Academic Service

SUPSI - Intelligent Energy Systems / ISAAC | Southern Switzerland | +39 334 612 9021 | roberto.rocchetta@supsi.ch



Research in uncertainty quantification, reliability, resilience, and data-driven decision-making for complex engineering systems, with applications in energy systems, infrastructure, and risk-informed operations.

Currently I hold a permanent research position at SUPSI. I have developed an internationally visible profile through collaborations with NASA Langley, NIA, TU Eindhoven, Philips Research, ETH Zurich, and Swiss partners including Swissgrid.

I have authored 30+ peer-reviewed publications, including 20+ journal articles, with about 1,200+ citations and h-index 15, and has been recognized through distinctions including Stanford's 2025 Top 2% ranking and an Alexander von Humboldt fellowship.

**RESEARCH OUTPUT**

## 30+ papers

40+ on Scopus; 20+ first-author journal articles; many transdisciplinary contributions

**SCHOLARLY IMPACT**

## ~1,300+ citations

Google Scholar h-index 16 and i10-index 18

**RECOGNITION**

## Top 2% researcher

Stanford 2025 world ranking; Humboldt fellowship

**APPLIED EXCELLENCE**

## Challenge-winning work

Best numerical solution in NASA-DNV 2025

**COMPETITIVE GRANTS**

## ~ 600+ k€

PROPER -Grids (275 kCHF); PEDAID (205 kCHF)
Hamburg-RL (98 k€); Alexander von Humboldt (60 k€)

**LANGUAGEs**

## Italian, English

Italian (mother tongue); English (IELTS & Cambridge Certificates); Spanish/French (B2 – B1); Dutch (A1);

**CONTACTS**


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## RESEARCH AGENDA

- **Uncertainty quantification**: forecasting, inverse problems, non-probabilistic methods, Approx Bayesian computation.
- **Decision-making under uncertainty**: Robust problems, scenario theory, distributionally robustness, risk informed.
- **Control and sequential decision-making**: reinforcement learning, reachability analysis, DRO, POMDPs.
- **Survival and resilience**: Risk assessment, reliability, resilience assessment and risk-informed decision-making.
- **Applications**: power, energy, health monitoring, structural, maintenance, mobility, prevention/recovery, planning.

## APPOINTMENTS AND EDUCATION

| Appointment | Institution / Focus | Period |
|---|---|---|
| **Researcher (permanent)** | SUPSI, Intelligent Energy Systems / ISAAC | Oct 2022 - Present |
| **Postdoctoral Researcher** | TU Eindhoven and Signify - stochastic methods, probability, and security | Oct 2019 - Oct 2022 |
| **Research Scholar** | National Institute of Aerospace and NASA Langley | Jan 2019 - Oct 2019 |
| **PhD Researcher** | University of Liverpool, Risk and Uncertainty Laboratory | 2015 - 2019 |

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|---|---|---|
| **Secondments and visiting** | ARAMIS srl, ETH Zurich, Politecnico di Milano, Ecole Centrale Paris | 2013 - 2018 |
| **PhD Degree** | Engineering Science and System Reliability, Uni Liverpool | 2015-2019 |
| **Master of Research** | Decision-Making under Risk and Uncertainty, Uni Liverpool | 2014-2015 |
| **Bachelor & Master's degree** | Energy Engineering, University of Bologna | 2008-2013 |

## SELECTED RESEARCH ACHIEVEMENTS

### High-Impact Research on Reinforcement Learning for Power Grid Maintenance

During my PhD, in collaboration with Prof. Dr. Patelli, Prof. Enrico Zio, and the Milan-based consultancy group formerly ARAMIS srl, I co-developed a pioneering reinforcement-learning framework for optimizing joint operation and maintenance strategies in power grids. The work integrated prognostics and health management with reinforcement learning to improve grid reliability and resilience, and was published in Applied Energy in 2019.

### Data-Driven Advances in Reliability-Based Design Optimization

In collaboration with Dr.-Ing. Luis G. Crespo and Dr.-Ing. Sean P. Kenny at NASA Langley Research Center, I helped develop a data-driven approach to reliability-based design optimization and risk-informed decision-making that avoids the traditional need to estimate uncertainty distributions. This publication series established the scenario-RBDO approach across controller design, broader RBDO problems, and extreme-risk settings.

### Independent and Collaborative Research in Resilience and Uncertainty Quantification

This publication series highlights both my ability to conduct independent research and to work effectively in multidisciplinary teams. It includes a sole-author paper in Renewable and Sustainable Energy Reviews on infrastructure resilience, a top-performing team response to NASA Langley's uncertainty quantification challenge developed during the COVID-19 lockdown, and a journal collaboration involving TU Eindhoven and Philips Research.

## TEACHING, MENTORING, & SUPERVISION

- **Lecturer**: Decision Making Under Uncertainty (2nd year USI - Bachelor in Data Science, 2025 and 2026) see ***link***
- **Lecturer**: Survival & Reliability Analysis (TU/Eindhoven, Bachelor program, 2021) see ***link***
- **Master thesis supervisor**: Berti D., 2025, ETHZ "*Risk-based Probabilistic Security Assessment: An Awareness Tool for Outage Validation at Swissgrid Operational Planning*" (confidential)
- **Master thesis supervisor**: Jongerius, S.C. 2022, TU/e, *Coverage of lifetime confidence bounds for highly censored, and few Weibull distributed data*, ***link***
- **Master thesis supervisor**: van der Putten, J. J. 2021, TU/e, *"Data mining in ECG data to predict patient deterioration in low resource settings"* ***link***
- **Teaching qualification**: SEDIFO introduction to didactic 2025

## SELECTED HONOURS & AWARDS

- ✓ **Stanford Top 2% ranking**, 2025 (security topic)
- ✓ **Reviewer of the Year**, ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems
- ✓ **Alexander von Humboldt** fellowship, 2022 (declined)
- ✓ **Best score, innovation challenge** *'Health management in changing environments'*, ESREL 2020
- ✓ **Best numerical solution, NASA-DNV challenge 2025** on optimization under uncertainty
- ✓ **6th place, CityLearn NeurIPS challenge 2024** on forecasting and optimization
- ✓ **Best poster award**, ISIPTA 2021
- ✓ **First prize,** "MC approach to compute the success probability of sending objects to the Moon", Math. Comp. Games 2016.

## PEER-REVIEWED JOURNAL PUBLICATIONS

[1] L Nespoli, A Biswas, **R. Rocchetta**, V Medici (2026) Nonlinear reconciliation: Error reduction theorems, Transaction on Machine Learning Research.
[2] P. Dersin, **R. Rocchetta** (2025). *Analysis of RUL dynamics and uncertainty via time transformation*, RESS, 111730 Reliability Engineering and System Safety. https://doi.org/10.1016/j.ress.2025.111730
[3] E. Congeduti, **R. Rocchetta**, and F. A. Oliehoek *Influence Learning in Complex Systems,* Transact on ML Research, 2025. pdf
[4] **R. Rocchetta**, L. Nespoli, V. Medici, A. Shemesh, Y. Parag, and J. M. Tardif *Optimization of Mobility Incentives in Electric Vehicle Car Sharing Systems: A Reinforcement Learning Framework,* Sustainable Cities & Society, Vol. 120, 2025.
[5] **R. Rocchetta**, E. Perrone, A. Herzog, P. Dersin, A. Di Bucchianico A survey on LED Prognostics and Health Management and uncertainty reduction, Microelectronics Reliability 157, 115399
[6] **R. Rocchetta**, Zhouzhao Zhan, Willem Dirk van Driel, Alessandro Di Bucchianico, "*Uncertainty analysis and interval prediction of LEDs lifetimes*", Reliability Engineering & System Safety, Vol 242, 2024, 109715, https://doi.org/10.1016/j.ress.2023.109715.
[7] **R. Rocchetta**, Alexander Mey, Frans Oliehoek. "*A Survey on Scenario Theory, Complexity, and Compression-Based Learning and Generalization*". IEEE Transactions on Neural Networks and Learning Systems, pp 1-15, 2023. https://doi.org/10.1109/TNNLS.2023.3308828
[8] Katharina Proksch, A. Di Bucchianico, **R. Rocchetta,** et al. "*Personalised Health Monitoring for Early Disease Detection",* Mathematics in Industry Reports, 2023, 10.33774/miir-2023-mgp7g
[9] Cangul, O., **R. Rocchetta**, Fahrioglu, M., Patelli, E., "*Optimal Allocation and Sizing of Decentralized Solar Photovoltaic Generators Using Unit Financial Impact Indicator*". *Sustainability* 2023, *15*, 11715. https://doi.org/10.3390/su151511715 **,**
[10] **R. Rocchetta**, Qi Gao, Dimitrios Mavroeidis, Milan Petkovic*, "A robust model selection framework for fault detection and system health monitoring with limited failure examples: Heterogeneous data fusion and formal sensitivity bounds*", Engineering Applications of Artificial Intelligence, Vol 114, 2022, 105140, https://doi.org/10.1016/j.engappai.2022.105140
[11] **R. Rocchetta**, "*Enhancing the resilience of critical infrastructures: a statistical analysis on spectral clustering and vulnerability metrics for power networks*", RSER, Vol 159, 2022, https://doi.org/10.1016/j.rser.2022.112185
[12] W. Zhao, C. Dang, **R. Rocchetta**, M. Valdebenito, D. Moens*, "Enriching stochastic model updating: An efficient Bayesian approach using the Bray-Curtis distance and an adaptive binning algorithm",* MSSP , 2022, https://doi.org/10.1016/j.ymssp.2022.108889,
[13] A. Gray, …, **R. Rocchetta**, *"From inference to design: A comprehensive framework for uncertainty quantification in engineering with limited information",* Mechanical Systems and Signal Processing. 165, 2022, 108210, https://doi.org/10.1016/j.ymssp.2021.108210
[14] **R. Rocchetta**, Qi Gao, Milan Petkovic, "*Soft-constrained interval predictor models and epistemic reliability intervals: a new tool for uncertainty quantification with limited experimental data",* MSSP, Vol 161, 2021, https://doi.org/10.1016/j.ymssp.2021.107973
[15] **R. Rocchetta**, *Luis G. Crespo, "A scenario optimization approach to reliability-based and risk-based design: soft-constrained modulation of failure probability bounds",* RESS, 2021, https://doi.org/10.1016/j.ress.2021.107900
[16] **R. Rocchetta**, E. Patelli, "*A Post-Contingency Power Flow Emulator for Generalized Probabilistic Risks Assessment of Power Grids",* RESS, Vol 197, May 2020, 106817, https://doi.org/10.1016/j.ress.2020.106817
[17] **R. Rocchetta**, Luis G. Crespo, Sean P. Kenny, "*A scenario optimization approach to reliability-based design*", RESS, Vol 196, 2020, https://doi.org/10.1016/j.ress.2019.106755
[18] **R. Rocchetta**, et al., "*A Reinforcement Learning Framework for Optimal Operation and Maintenance of Power Grids",* Applied Energy, Vol 241, Pp 291-301, 2019, https://doi.org/10.1016/j.apenergy.2019.03.027
[19] **R. Rocchetta**, E. Patelli, E. Zio, "*A Power-Flow Emulator Approach for Resilience Assessment of Repairable Power Grids subject to Weather-Induced Failures and Data Deficiency",* Applied Energy, Vol 210, 2018, https://doi.org/10.1016/j.apenergy.2017.10.126
[20] **R. Rocchetta**, E. Patelli, "*Assessment of Power Grid Vulnerabilities Accounting for Stochastic Loads and Model Imprecision*", IJEPES, Vol 98, pp 219-232, 2018, https://doi.org/10.1016/j.ijepes.2017.11.047
[21] **R. Rocchetta,** E. Patelli, M. Broggi, Q. Huchet, *"On-Line Bayesian Model Updating for Structural Health Monitoring",* MSSP, Vol 103, 174 - 195, 2018, https://doi.org/10.1016/j.ymssp.2017.10.015
[22] **R. Rocchetta**, E. Patelli, M. Broggi*, "Do we have enough data? Robust reliability via uncertainty quantification*", Applied Mathematical Modelling, Volume 54, pp 710-721, 2018, https://doi.org/10.1016/j.apm.2017.10.020
[23] **R. Rocchetta,** Y.F. Li and E. Zio, "*Risk Assessment and Risk-Cost Optimization of Distributed Generation Systems Considering Extreme Weather Conditions",* RESS, Vol 136, pp 47 - 61, 2015, https://doi.org/10.1016/j.ress.2014.11.013

## PEER-REVIEWED CONFERENE PUBLICATIONS

[1] Davide Berti, Davood Raoofsheibani, Blazhe Gjorgiev, **R. Rocchetta**, Giovanni Sansavini *A Probabilistic Risk Assessment Framework to Estimate Power System Risks for Outage Planning*, Power Systems Computational Conference, 2026.
[2] **R. Rocchetta**, L. Nespoli, V. Medici, Y. Chen, M. de Angelis, D. Ochnio, E. Patelli, E. Smith, "*An Integrated Uncertainty Quantification and Optimization for solving the 2025 NASA-DNV Challenge*" ESREL conference 2025
[3] **R. Rocchetta**, B. Gjorgiev, A. Bellè, "Towards Risk-Informed Transmission Grid Outage Planning", ESREL conference, 2025
[4] **R. Rocchetta**, L. Nespoli, V. Medici, D. Raoofsheibani, B. Gjorgiev, G. Sansavini*, "Risk Informed Operational Planning of Power Transmission Grids: Overview of Recent Developments",* ESREL 2025
[5] **R. Rocchetta,** L. Nespoli, V. Medici, S. Basso, M. Derboni, M. Salani, "*Rule-based deep reinforcement learning for optimal control of electrical batteries in an energy community ",* ESREL 2023 *link*
[6] **R. Rocchetta,** Z. Zhao, A. Di Bucchianico, "*Prediction of the Luminous Flux Degradation of Light Emitting Diodes with an Interval Regressions Model ",* ESREL 2022 *link*
[7] Wang, Chenxing, Lechang Yang, **R. Rocchetta**, *"Bayesian Information Fusion for Imprecise Probabilistic Models with Different Types of Information."* 3rd International Conference on System Reliability and Safety Engineering (SRSE). IEEE, 2021.
[8] **R. Rocchetta,** "*New probabilistic guarantees on the accuracy of Extreme Learning Machines: an application to decision-making in a reliability context",* ESREL conference, September 2021 link
[9] M. De Angelis, **R. Rocchetta**, A. Gray, S. Ferson, *"Constructing consonant predictive beliefs from data with scenario theory*", International Symposium on Imprecise Probabilities ISIPTA: Theories and Applications*, July 2021 link*
[10] A. Gray, A. Wimbush, **R. Rocchetta**, M. De Angelis, P. O. Hristov, E. Miralles-Dolz, D. Calleja, "*Bayesian calibration and probability bounds analysis: solution to the Nasa 2020 UQ challenge on optimization under uncertainty",* ESREL-PSAM Conferences, 2020 link
[11] **R. Rocchetta**, M. Petkovic, Q. Gao, "*Scenario-based Generalization bound for Anomaly Detection Support Vector Machine Ensembles*" , Proceedings of the 30th ESREL and the 15th PSAM Conferences, 2020 link
[12] **R. Rocchetta**, L. G. Crespo, "*An empirical approach to reliability-based design using scenario optimization",* Proceedings of the 30th ESREL and the 15th PSAM Conferences, 2020 link
[13] **R. Rocchetta**, L. G. Crespo, S. P Kenny *"Solution of the benchmark control problem by scenario optimization"* ASME 2019 Dynamic Systems and Control Conference, DSCC , link

[14] E. Patelli, S. Tolo, H. George-Williams, J. Sadeghi, **R. Rocchetta**, M. de Angelis, M. Broggi *"OpenCossan 2.0: an efficient computational toolbox for risk, reliability and resilience analysis",* Proceedings of the joint ICVRAM ISUMA Uncertaintes conference, 2018. link
[15] **R. Rocchetta**, M. Compare, E. Patelli, L. Bellani, E. Zio, *"A reinforcement learning framework for optimisation of power grid operations and maintenance",* 8th international workshop on reliable engineering computing, REC 2018, Liverpool, UK, Jully, 2018. link
[16] **R. Rocchetta**, E. Patelli, *"Stochastic Analysis and Reliability-Cost Optimization of Distributed Generators and Air Source Heat Pumps",* International Conference on System Reliability and Safety, ICSRS, 2017, link
[17] **R. Rocchetta**, E. Patelli, *"An Efficient Framework for Reliability Assessment of Power Networks Installing Renewable Generators and Subject to Parametric P-box Uncertainty",* Proceedings of the 27th ESREL conference, link
[18] **R. Rocchetta**, E. Patelli, "*Power Grid Robustness to Severe Failures: Topological and Flow Based Metrics Comparison",* European Congress on Computational Methods in Applied Sciences and Engineering, ECCOMAS, Crete 2016, pp. 6121-6135. link
[19] **R. Rocchetta**, E. Patelli, M. Broggi, Q. Huchet, *"On Bayesian Approaches for Real-Time Crack Detection",* ESREL conference proceedings, Zurich 2015, pp 1929-1936. link
[20] **R. Rocchetta**, E. Patelli, "*Imprecise Probabilistic Framework for Power Grids Risk Assessment and Sensitivity Analysis",* European Safety and Reliability Conference, ESREL, Glasgow 2016.
[21] **R. Rocchetta**, E. Patelli, M. Broggi, *"Efficient Epistemic-Aleatory Uncertainty Quantification: Application to the NAFEMS Challenge Problem",* NAFEMS World Congress, San Diego, California, USA 21-24 June 2015. link
[22] **R. Rocchetta**, E. Patelli, *"A Simulation-Based Probabilistic Risk Assessment of Electric Vehicles Control Strategies Accounting Renewable Energy Sources",* International Probabilistic Workshop, IPW Liverpool, UK, 4-6 November 2015, pp. 183-198.

## REFEREES AND NETWORK

1. **Prof. Dr. Edoardo Patelli**, Strathclyde university, former Ph.D, supervisor edoardo.patelli@strath.ac.uk
2. **Prof. Dr. Ing- Enrico Zio**, Polytechnic Milan and Mines ParisTech, academic advisor, enrico.zio@poilimi.it
3. **Prof. Dr. Pierre Dersin**, Lulea university, accademic collaborator, pierre.dersin@ltu.se
4. **Prof. Dr. Milan Petkovic,** project coordinator at TU/e and Philips, m.petkovic@tue.nl
5. **Late Prof Alessandro Di Bucchianico,** TU/Eindhoven (contact Elisa Perrone as reference e.perrone@tue.nl)
6. **Prof. Dr. Ing- Giovanni Sansavini**, ETH Zurich, sansavig@ethz.ch
7. **Dr. Luis G. Crespo,** former collaborator at NASA Langley Research Center, luis.g.crespo@nasa.gov
8. **Prof. Dr. Ing- Michael Beer**, Leibniz University Hannover, beer@bauinf.uni-hannover.de
9. **Prof Dr. Scott Ferson**, Chair in risk and uncertainty, Liverpool, Scott.Ferson@liverpool.ac.uk

## SOFTWARE and Tools

- **Programming languages**: MATLAB, Python, Julia, R-Studio, GitHub, Docker
- **Machine learning libraries**: TensorFlow, Stable-Baseline, Keras
- **Data base, data management**: SQL, Rabbit
- **Writing and Visualization**: Overleaf, Beamer, Office, Inkscape.
- **Project Management and Productivity:** Asana, Notion, Slack, Microsoft Teams, Codex, Perplexity

## PEER RECOGNITION AND STANDING IN THE RESEARCH COMMUNITY

My research standing is reflected not only in publications, but also in recognition by peers and the wider research community. According to my current academic record, I have authored over 30 peer-reviewed publications, including more than 20 journal articles, with approximately 1,000-1,300 citations and an h-index of 15. My work has been recognized through distinctions including Stanford's 2025 Top 2% ranking, the Alexander von Humboldt fellowship, the NASA-DNV 2025 best numerical solution award, and Reviewer of the Year recognition from the ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems. My standing is also reflected in invitations to serve as keynote speaker, session chair, special-session co-organizer, topic board editor, and reviewer for leading journals in reliability, risk, energy systems, and uncertainty quantification.

## ADDITIONAL SELECTION OF PROFESSIONAL SERVICES & ACTIVITIES

- Special session co-organizer and co-chair, Hybrid PHM, ESREL 2023
- Invited keynote session, ICSRS 2022 and 2023
- Invited speaker, Julia Reach / Julia Interval workshop (2021)
- Topic Board Editor, Mathematics (MDPI)
- Session Chair, uncertain systems and robustness, DSCC 2019
- Reviewer for IJEPES, RESS, MSSP,
- Reviewer for Risk Analysis, IET GTD, Mathematics, Energies,
- Reviewer for ASCE-ASME Journal of Risk and Uncertainty

